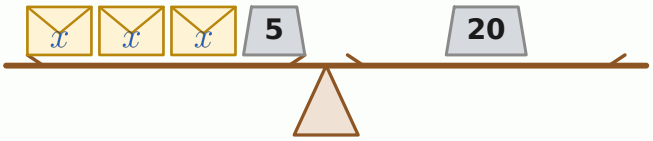
Chapter 15

Solving equations

finding the number hiding in the envelope



three sealed envelopes and a 5 g weight balance a 20 g weight. How heavy is one envelope?

$$3x + 5 = 20$$

An **equation** says two things are equal: the scales balance. **Solving** it means finding the number (or numbers) that the letter must be for the balance to hold. That number is called the **solution**.

Every move in this chapter is one of the "undo buttons" from Chapter 14, done to **both sides at once** so the scales stay level. After one letter, you'll meet two letters at once (café receipts) and squared letters, where Chapter 12's pair turns straight into the quadratic formula. Every answer gets put back into the question as a check.

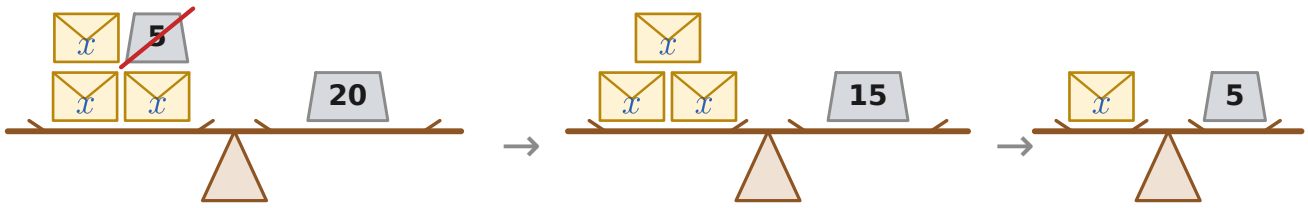
Goes with **Lecture 2** (rearranging), **Lecture 3** (mixing solutions, compounds, simultaneous equations) and **Lecture 14** (solving quadratics).

Balance scales: do the same to both pans

PROCEDURE: solving a linear equation

1. Clear **brackets** (expand) and **fractions** (multiply every term by the LCD).
2. Collect the letter terms on one side (the side with more of them) and the numbers on the other, doing the same to both sides.
3. Divide both sides by the number in front of the letter.
4. **Check:** put the answer back into the original equation.

You could solve $3x + 5 = 20$ by trying numbers until both sides match, but that fails for answers like $x = 1.875$. The no-guessing way: each move keeps the scales level, so whatever you take from one pan, you take from the other:



take 5 from both *3 envelopes = 15* *one = 5*

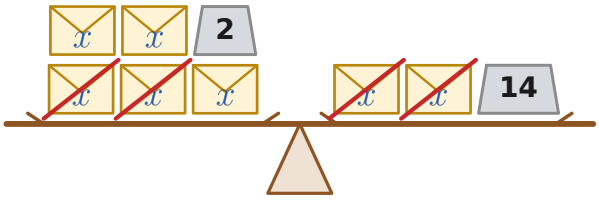
$$3x + 5 = 20$$

$$3x = 15 \quad \longleftarrow \text{–5 on both sides}$$

$$x = 5 \quad \longleftarrow \text{÷3 on both sides (share into 3 equal groups)}$$

Always check: put the answer back. $3(5) + 5 = 20$ ✓.

Letters on both sides: $5x + 2 = 2x + 14$. Take the **same number of envelopes** off each pan, just as you'd take weights off:



take 2 envelopes from each pan

2 envelopes crossed out on EACH pan: what's left still balances

$$5x + 2 = 2x + 14$$

$$3x + 2 = 14 \quad \longleftarrow \text{–2x on both sides (2 envelopes off each pan)}$$

$$3x = 12 \;\Rightarrow\; x = 4$$ ← −2, then ÷3

Check: left $5(4) + 2 = 22$, right $2(4) + 14 = 22$ ✓. *Tip:* move the letters to whichever side has **more** of them, so you never end up with a negative number of envelopes.

Brackets and fractions: tidy first, then balance

Brackets: expand them (Chapter 11), then it's an ordinary balance. $3(x - 2) = 2(x + 1)$:

$$3x - 6 = 2x + 2$$ ← expand both sides

$$x - 6 = 2$$ ← −2x

$$x = 8$$ ← +6

Check: $3(6) = 18$ and $2(9) = 18$ ✓.

Fractions: multiply **both sides** by something that clears every bottom. For $\frac{x+1}{3} = \frac{x-1}{2}$, use 6 (both 3 and 2 go into it):

$$6 \times \frac{x+1}{3} = 6 \times \frac{x-1}{2}$$ ← × 6 on both sides

$$2(x+1) = 3(x-1)$$ ← 6 ÷ 3 = 2; 6 ÷ 2 = 3

$$2x + 2 = 3x - 3 \;\Rightarrow\; x = 5$$ ← −2x and +3 on both sides

Look at step 2: each top ended up multiplied by the **other** side's bottom. That shortcut has a name.

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; ad = cb$$

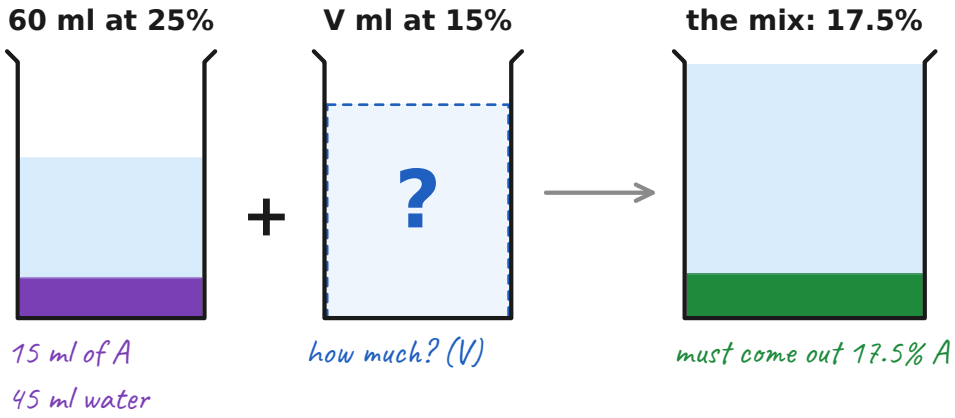
"cross multiplying" is just × b and × d on both sides, done in one go

The Chapter 4 promise: Lecture 3 turns $\frac{5}{V} = \frac{8}{3}$ into $15 = 8V$. That's cross multiplying: $5 \times 3 = 8 \times V$, so $V = \frac{15}{8} = 1.875$ m/s.

Lecture 3, Mixing 2 and 3: an equation from beakers

How much 15% solution must be added to 60 ml of 25% solution to make a 17.5% solution? Before any maths, let's decode what that sentence is asking.

What "a 25% solution" means. A solution is a chemical (call it A) dissolved in water. "25%" says a quarter of the liquid is A and the rest is water. In the pictures below, imagine the A could be poured out separately: the dark part is A, the pale part is water.



In plain English: you have a strong mix. You pour in a weaker one, which dilutes it. Pour a little and it's still strong; pour loads and it's nearly down to 15%. Somewhere in between it is exactly 17.5%. **How much do you pour to stop right there?** That amount is V.

To find it, **follow the chemical A**, not the percentages. Count how much A is in each beaker:

60 ml at 25%	+	**V ml at 15%**	=	**60 + V ml together**
amount of A: $0.25 \times 60 = 15$		amount of A: $0.15V$		amount of A: $15 + 0.15V$

percentage of the mix = (amount of A) ÷ (total volume) × 100, and it must be 17.5

$$\frac{15 + 0.15V}{60 + V} \times 100 = 17.5$$ ← *the equation (the slide's)*

$$(15 + 0.15V) \times 100 = 17.5(60 + V)$$ ← *× (60 + V) on both sides*

$$1500 + 15V = 1050 + 17.5V$$ ← *expand both sides*

$$450 = 2.5V \quad \longleftarrow \text{ } -1050 \text{ and } -15V \text{ on both sides}$$

$$V = \frac{450}{2.5} = 180 \quad \longleftarrow \text{ } \div 2.5$$

Check with the beakers: A = $15 + 0.15 \times 180 = 15 + 27 = 42$ ml in $60 + 180 = 240$ ml. $\frac{42}{240} \times 100 = 17.5\%$ ✓. So add **180 ml** of the 15% solution.

Two unknowns: café receipts

You find two receipts. You don't know the price of a coffee (c) or a cake (k):

CAFÉ	
coffee	
coffee	
cake	
TOTAL	**£7.00**

CAFÉ	
coffee	
cake	
TOTAL	**£4.50**

Put them side by side. The first has exactly ONE more coffee than the second, and costs £2.50 more. So a coffee is £2.50!

That's the whole idea of **simultaneous equations**: two facts about the same two unknowns, used together. Written in algebra:

$$2c + k = 7 \qquad (1)$$

$$c + k = 4.5 \qquad (2)$$

$$c = 2.5 \quad \longleftarrow \text{ } (1) - (2)\text{: the cakes cancel}$$

$$2.5 + k = 4.5 \;\Rightarrow\; k = 2 \quad \longleftarrow \text{ put } c \text{ into } (2)$$

Check in the other one: $2(2.50) + 2 = 7$ ✓. Subtracting whole equations like this is called **elimination**. Here is the rule in full:

PROCEDURE: elimination

1. Label the equations (1) and (2), lined up: x under x, y under y, numbers on the right.
2. Multiply one equation, **or both**, by numbers that make the coefficients of one letter the **same size**.
3. **Same signs: subtract** the equations. **Opposite signs: add** them. That letter cancels.
4. Solve for the letter that's left.
5. Put it back into either original equation to find the other letter. Check in the other one.

Scale one equation (three copies of a receipt cost three times as much):

$$3x + 2y = 16$$

$$x + y = 6$$

$$2x + 2y = 12 \longleftarrow \times 2 \text{ the second equation}$$

$$x = 4 \longleftarrow +2y \text{ in both (same sign): subtract}$$

$$4 + y = 6 \;\Rightarrow\; y = 2 \longleftarrow \text{put } x = 4 \text{ into } x + y = 6$$

Scale both equations: $3x + 2y = 12$ (1) and $4x - 3y = -1$ (2). No single multiplier makes anything match, so aim for $6y$ in both:

$$9x + 6y = 36 \qquad (1) \times 3$$

$$8x - 6y = -2 \qquad (2) \times 2$$

$$17x = 34 \;\Rightarrow\; x = 2 \longleftarrow +6y \text{ and } -6y \text{ (opposite signs): add}$$

$$6 + 2y = 12 \;\Rightarrow\; y = 3 \longleftarrow x = 2 \text{ into (1)}$$

Check in (2): $4(2) - 3(3) = 8 - 9 = -1$ ✓.

The substitution method

PROCEDURE: substitution

1. **Rearrange** one equation to get one letter on its own (pick a letter with no number in front).
2. **Substitute** that expression into the **other** equation, in brackets.
3. Solve the one-letter equation you now have.
4. **Back-substitute**: put the answer into the rearranged equation to find the other letter.
5. Check both answers in **both** original equations.

Example: $2x - y = 1$ (1) and $3x + 2y = 12$ (2).

$$y = 2x - 1$$ ← 1. rearrange (1): y has no number in front

$$3x + 2(2x - 1) = 12$$ ← 2. substitute into (2)

$$3x + 4x - 2 = 12 \;\Rightarrow\; 7x = 14 \;\Rightarrow\; x = 2$$

3. solve

$$y = 2(2) - 1 = 3$$ ← 4. back-substitute

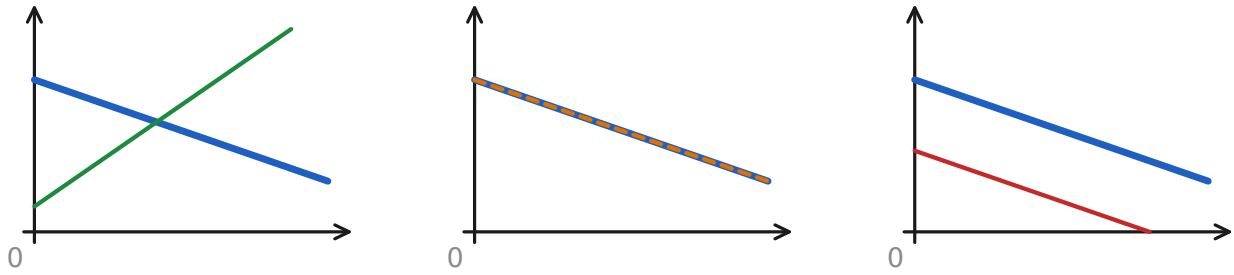
5. Check: (1) $4 - 3 = 1$ ✓, (2) $6 + 6 = 12$ ✓.

Which method? Substitution is quickest when a letter is already on its own (like $y = 2x - 1$ or $x = 1000 - y$). Elimination is quickest when the equations are lined up with whole numbers. Both always give the same answer, and Lecture 12 (matrices) solves the same systems a third way.

The picture: where two lines cross

Each equation on its own has endless answers. $c + k = 4.5$ could be (1, 3.5), (2, 2.5), (3, 1.5)... Plot them all and you get a straight line (Chapter 20). The other receipt gives another line, and the answer is the one point on **both**: where they cross, at (2.5, 2).

Two lines can meet in three ways, and that decides how many answers there are:



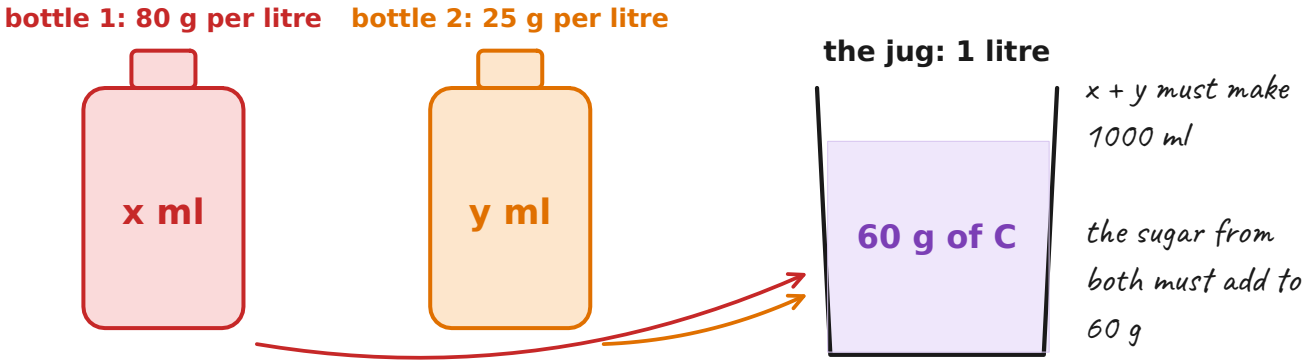
cross once: one answer **same line: endless answers** **parallel: no answer**

Lecture 3 ends with the middle case: $x + 2y = 10$ and $3x + 6y = 30$. The second is just the first multiplied by 3 (the same order bought three times), so it tells you nothing new. Try to solve and everything cancels to $30 = 30$: true, but no letter left to find. These equations are **not independent**. Two equations that contradict each other, like $x + 2y = 10$ and $x + 2y = 12$, are the parallel case: no answer at all.

Lecture 3 compounds: two receipts in disguise

Solution 1 has 80 g of C per litre, solution 2 has 25 g per litre. How much of each makes 1 litre with 60 g per litre?

Decoding it. Think of two bottles of squash, with C as the sugar. "80 g per litre" means every litre of bottle 1 has 80 g of sugar dissolved in it: strong. Bottle 2 has only 25 g per litre: weak. You need to fill a **1-litre jug** that contains exactly **60 g** of sugar. **How many ml do you pour from each bottle?** That's two unknowns.



Every ml of bottle 1 carries $80 \div 1000 = 0.08$ g; every ml of bottle 2 carries 0.025 g. So the two facts are two receipts:

$$x + y = 1000 \qquad (1)$$ ← the volumes add to 1 litre

$$0.08x + 0.025y = 60 \qquad (2)$$ ← the grams of C add to 60

Elimination, exactly as with the café: make the x parts match, then subtract.

$$0.08x + 0.08y = 80$$ ← (1) × 0.08

$$0.055y = 20 \;\Rightarrow\; y = \frac{20}{0.055} \approx 364$$ ← this − (2): the x's cancel

$$x = 1000 - 364 = 636$$ ← back into (1)

Check: $0.08 \times 636 + 0.025 \times 364 = 50.9 + 9.1 = 60$ g ✓. The slide uses **substitution** instead (rearrange (1) to $x = 1000 - y$ and put that into (2): $0.08(1000 - y) + 0.025y = 60$, so $80 - 0.055y = 60$); it gives the same answer. The slide prints $\frac{20}{0.55}$; it should be $\frac{20}{0.055}$, as the answer 364 shows.

Squared letters: two answers, and the zero trick

RULE: the zero product rule

If $A \times B = 0$, then $A = 0$ or $B = 0$. This works **only for 0**.

So **rearrange so one side is 0 BEFORE you factorise**. $(x-1)(x-2) = 6$ does **not** give $x - 1 = 6$: two numbers can multiply to 6 in endless ways.

PROCEDURE: solving a quadratic equation

1. **Rearrange to** $ax^2 + bx + c = 0$: expand any brackets, move every term to one side.
2. **Factorise** (Chapter 12). If it won't factorise, use the **quadratic formula** (next section).
3. Set **each bracket equal to 0** and solve each one.
4. **Check** each answer in the original equation, and reject any that don't fit the question.

$x^2 = 25$ has **two** solutions: $5 \times 5 = 25$ and also $(-5) \times (-5) = 25$ (Chapter 9 warned you). We write $x = \pm 5$, said "plus or minus five".

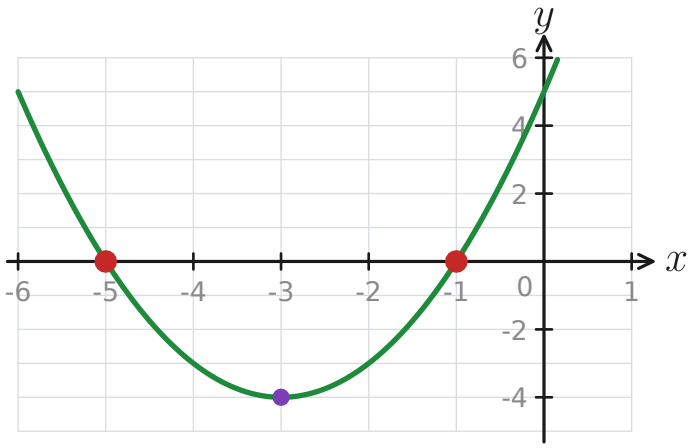
An equation with an x^2 is a **quadratic equation**. Lecture 14's example: $x^2 + 6x + 5 = 0$. The key fact is the **zero trick**: if two numbers multiply to give 0, one of them must be 0 (there's no other way to make 0 by multiplying).

$$x^2 + 6x + 5 = 0$$

$$(x+5)(x+1) = 0 \longleftarrow \text{factorise (Chapter 12: } \Delta = 16\text{, pair 5 and 1)}$$

$$x + 5 = 0 \quad \text{or} \quad x + 1 = 0 \longleftarrow \text{one bracket must be zero}$$

$$x = -5 \quad \text{or} \quad x = -1$$



The graph of $y = x^2 + 6x + 5$ is a U-shaped curve. The solutions are where it crosses the x-axis (where $y = 0$): $x = -5$ and $x = -1$.

The bottom of the U, $(-3, -4)$, is the vertex (two sections on).

Not zero yet: $(x-1)(x-2) = 6$. Expand and rearrange first:

$$x^2 - 3x + 2 = 6$$ ← *expand*

$$x^2 - 3x - 4 = 0$$ ← *−6 on both sides: now one side is 0*

$$(x-4)(x+1) = 0 \Rightarrow x = 4 \text{ or } x = -1$$ ← *factorise; each bracket = 0*

Check: $(3)(2) = 6$ ✓ and $(-2)(-3) = 6$ ✓.

✗ Common mistakes

✗ $(x-1)(x-2) = 6 \Rightarrow x - 1 = 6$

the zero trick only works when the other side is 0

✓ $x^2 - 3x - 4 = 0 \Rightarrow x = 4,\ -1$

✗ $x^2 = 3x \Rightarrow x = 3$

dividing by x loses x = 0: factorise instead

✓ $x(x-3) = 0 \Rightarrow x = 0,\ 3$

✗ $x^2 = 25 \Rightarrow x = 5$

a square root gives two answers

✓ $x = \pm 5$

The quadratic formula: straight from Chapter 12

Chapter 12 factorises **any** quadratic with its pair $m, n = \frac{b \pm \sqrt{\Delta}}{2}$, where $\Delta = b^2 - 4ac$. Chapter 12's grid formula gives the brackets. Now use the zero trick on them:

$$ax^2 + bx + c = \left(x + \frac{n}{a}\right)(ax + m) = 0$$ ← Chapter 12's brackets

$$x = -\frac{n}{a} \quad \text{or} \quad x = -\frac{m}{a}$$ ← one bracket must be zero

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$ ← put in m, n = (b ± √Δ)/2

That's the famous **quadratic formula**. It is Chapter 12's pair with the signs flipped and divided by a: nothing new to learn. Treat it like a recipe card:

RECIPE: solving ax² + bx + c = 0

Ingredients:

a = number in front of x^2
b = number in front of x
c = the plain number

Method:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Lecture 14's example: $x^2 + 3x + 2 = 0$. Ingredients: $a = 1$, $b = 3$, $c = 2$. Work out Δ first, as in Chapter 12:

$$\Delta = 3^2 - 4 \times 1 \times 2 = 1$$ ← a perfect square: it would factorise too

$$x = \frac{-3 \pm 1}{2} = -1 \quad \text{or} \quad -2$$ ← + version, then − version

Where it earns its keep: $2x^2 - 4x - 3 = 0$. $a = 2$, $b = -4$, $c = -3$. **Careful:** $-b$ is $-(-4) = +4$.

$$\Delta = (-4)^2 - 4 \times 2 \times (-3) = 16 + 24 = 40$$

not a square: no neat brackets

$$x = \frac{4 \pm \sqrt{40}}{4} \approx 2.58 \quad \text{or} \quad -0.58 \leftarrow \sqrt{40} \approx 6.32$$

So: Δ a perfect square, factorising and the formula both work. Not a square, the formula is the quick route.

Completing the square: where the U bottoms out

RULE: completing the square

$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c.$

In words: **halve the number in front of x**, put it in a squared bracket, then **subtract that half squared** (the bracket adds it on, so take it off again) and keep c.

If $a \neq 1$: take a out of the x^2 and x terms first, complete the square inside, then multiply the a back in.

The result $a(x+p)^2 + q$ is **vertex form**: the vertex (turning point) is at $(-p,\ q)$.

It's worth knowing because it shows the bottom of the U. Lecture 14's example:

$$x^2 + 6x + 5 = (x+3)^2 - 9 + 5 = (x+3)^2 - 4$$

half of 6 is 3; $3^2 = 9$

When a is not 1: $2x^2 + 12x + 7$ (Lecture 14 uses this vertex form).

$$= 2(x^2 + 6x) + 7 \leftarrow \text{take 2 out of the x terms only}$$

$$= 2\left[(x+3)^2 - 9\right] + 7 \leftarrow \text{complete the square inside}$$

$$= 2(x+3)^2 - 18 + 7 = 2(x+3)^2 - 11 \leftarrow \text{multiply the 2 back in}$$

Vertex $(-3, -11)$. **Check** $x = 0$: 7 and $2(9) - 11 = 7$ ✓.

Solving by completing the square: $x^2 + 4x - 1 = 0$ won't factorise, but:

$$(x+2)^2 - 4 - 1 = 0 \ \Rightarrow\ (x+2)^2 = 5 \leftarrow \text{half of 4 is 2}$$

$$x + 2 = \pm\sqrt{5} \;\Rightarrow\; x = -2 \pm \sqrt{5}$$ ← square root: ±

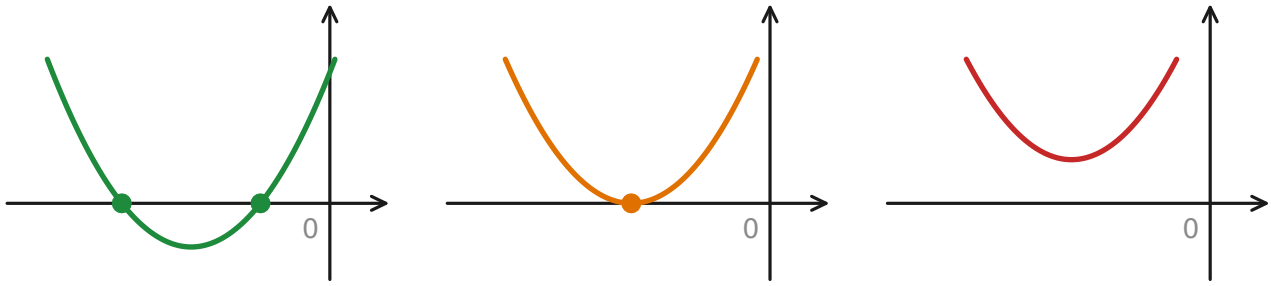
A square is never negative, so $(x+3)^2 - 4$ is smallest when the bracket is 0: at $x = -3$, where it's -4. That point, $(-3, -4)$, is the **vertex**. It also solves the equation by pure undoing: $(x+3)^2 = 4$, so $x + 3 = \pm 2$ and $x = -1$ or -5, the same answers as before.

How many answers? Look at Δ

The part under the root, $\Delta = b^2 - 4ac$, is Chapter 12's test number (the **discriminant**). It decides the brackets there, and the number of answers here. Take three neighbours that differ only in the last number:

✎ YOUR GO (pencil first)

$y = x^2 + 4x + 3$, $y = x^2 + 4x + 4$ and $y = x^2 + 4x + 5$ are three U-shaped graphs. Work out Δ for each, then predict how many times each U crosses the x-axis (where $y = 0$) before you look below.



$x^2 + 4x + 3 :\ \Delta = 4$ — **two crossings**

$x^2 + 4x + 4 :\ \Delta = 0$ — **one touch**

$x^2 + 4x + 5 :\ \Delta = -4$ — **no crossings**

Positive: the formula's $\pm$ gives two different answers, and the U crosses twice. Zero: $+0$ and -0 are the same, so one answer, and the U just touches. Negative: there is no real square root (Chapter 9), so no **real** answers, and the U floats above (or below) the axis. Chapter 12's table of cases is the same fact.

$\Delta > 0$	$\Delta = 0$	$\Delta < 0$
two answers	**one answer**	**no real answers**

Find the unknown: partial information about a quadratic

Exam questions often hide a letter (usually k) in the equation and give you **one fact** about the answers. Each fact turns into an equation for k:

RULE: what each fact means

"**Equal roots**" (or "one repeated root", "touches the x-axis"): $\Delta = b^2 - 4ac = 0$.

"**Two different real roots**": $\Delta > 0$. "**No real roots**": $\Delta < 0$.

"**x = r is a root**": put $x = r$ into the equation; it must come out true. Solve for k.

Then answer the rest (e.g. the other root) by solving the now-complete equation.

Example 1: find k so that $x^2 + kx + 9 = 0$ has equal roots.

$$\Delta = k^2 - 4 \times 1 \times 9 = k^2 - 36$$ ← a = 1, b = k, c = 9

$$k^2 - 36 = 0 \;\Rightarrow\; k^2 = 36 \;\Rightarrow\; k = \pm 6$$ ← equal roots: Δ = 0

Check $k = 6$: $x^2 + 6x + 9 = (x+3)^2$, one repeated root $x = -3$ ✓. ($k = -6$ gives $(x-3)^2$.)

Example 2: one root of $x^2 + kx - 10 = 0$ is $x = 2$. Find k and the other root.

$$2^2 + 2k - 10 = 0$$ ← x = 2 makes it true

$$2k - 6 = 0 \;\Rightarrow\; k = 3$$ ← solve for k

$$x^2 + 3x - 10 = (x-2)(x+5) = 0$$ ← now factorise the full equation

$$\text{other root: } x = -5$$

Check: $(-5)^2 + 3(-5) - 10 = 25 - 15 - 10 = 0$ ✓. (The $(x-2)$ bracket had to be there, because $x = 2$ is a root.)

Quadratics from words, and x on the bottom

RULE: rejecting answers

After solving, test every answer against the **question**: a length, time, mass or count can't be negative, so reject a negative answer there (say why).

Reject any answer that makes a **denominator 0** in the original equation.

Word problem: a rectangle is 3 cm longer than it is wide, and its area is 40 cm². Find its size.

$$x(x+3) = 40$$ ← width x, length $x+3$; area = width × length

$$x^2 + 3x - 40 = 0$$ ← expand, then rearrange to 0

$$(x+8)(x-5) = 0 \;\Rightarrow\; x = -8 \text{ or } x = 5$$ ← pair 8, −5

$$x = 5 : \; 5 \text{ cm by } 8 \text{ cm}$$ ← reject −8: a width can't be negative

Check: $5 \times 8 = 40$ ✓, and 8 is 3 more than 5 ✓.

x on the bottom: $x + \frac{6}{x} = 5$. Banned: $x = 0$. Multiply every term by x (Chapter 13):

$$x^2 + 6 = 5x \;\Rightarrow\; x^2 - 5x + 6 = 0$$ ← × x, then rearrange to 0

$$(x-2)(x-3) = 0 \;\Rightarrow\; x = 2 \text{ or } 3$$ ← neither is banned

Check: $2 + 3 = 5$ ✓ and $3 + 2 = 5$ ✓.

A banned answer: $\frac{x^2}{x-1} = \frac{1}{x-1}$. Banned: $x = 1$.

$$x^2 = 1 \;\Rightarrow\; x = \pm 1$$ ← × (x − 1)

$$x = -1 \text{ only}$$ ← x = 1 makes the bottom 0: reject it

Check: $\frac{1}{-2} = \frac{1}{-2}$ ✓.

Inequalities: solve like an equation, with one twist

RULE: solving an inequality

Use the same moves as for an equation: add or subtract anything on both sides; multiply or divide both sides by a **positive** number. The sign stays the same.

Multiplying or dividing both sides by a NEGATIVE number flips the sign: $<$ becomes $>$, $\leq$ becomes $\geq$ (and the other way round).

The answer is a **range** of values. Show it on a number line: **open circle** for $<$ or $>$ (the end is not included), **filled circle** for $\leq$ or $\geq$ (it is).

Check with one number from inside the range and one from outside.

Why the flip? $2 < 5$ is true. Multiply both sides by -1: -2 and -5. Now -2 is the **bigger** one, so $-2 > -5$. Multiplying by a negative reverses the order of numbers.

Example 1: $3x + 2 < 14$.

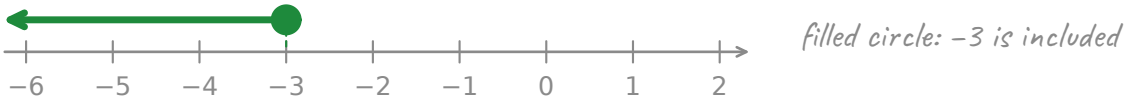
$$3x < 12 \;\Rightarrow\; x < 4$$ ← −2, then ÷3 (positive: no flip)



Example 2: $5 - 2x \geq 11$.

$$-2x \geq 6$$ ← −5 on both sides

$$x \leq -3$$ ← ÷ (−2): FLIP ≥ to ≤



Check: $x = -4$ (inside): $5 + 8 = 13 \geq 11$ ✓. $x = 0$ (outside): $5 \geq 11$ is false ✓.

Example 3 (two-sided): $-1 < 2x + 3 \leq 9$. Do each move to **all three** parts:

$$-4 < 2x \leq 6 \;\Rightarrow\; -2 < x \leq 3$$ ← −3, then ÷2

Solving rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Balance	do the same to both sides; check the answer in the original
Linear equations	clear brackets and fractions, letters to one side, divide, check
Cross-multiplying	$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$
Elimination	scale one or both equations to match a letter; **same signs subtract, opposite signs add**
Substitution	rearrange one equation for a letter, substitute into the other, solve, back-substitute, check both
Zero product	$AB = 0 \Rightarrow A = 0$ or $B = 0$ **rearrange to = 0 before factorising**
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$
Discriminant	$\Delta = b^2 - 4ac$ > 0 two roots, $= 0$ equal roots, < 0 no real roots
Given a root	substitute it; it must make the equation true
Completing the square	$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$ $a \neq 1$: take a out of the x terms first; vertex at $(-p, q)$
Rejecting answers	negative lengths/times; values making a denominator 0
Inequalities	as equations, but **× or ÷ by a negative flips the sign**; open circle $<$, filled $\leq$

Balance scales: your turn

Answers in grey. Put every answer back into the question to check it.

One unknown

1. $4x - 7 = 13$
2. $6x + 1 = 2x + 17$
3. $5(x - 3) = 2x + 3$
4. $(x + 4)/3 = (x - 2)/2$
5. How much 10% solution must be added to 40 ml of 30% solution to make 20%?

1. $x = 5$ 2. $x = 4$ 3. $x = 6$ 4. $x = 14$ 5. 40 ml

Two unknowns

6. $x + y = 10,\ x - y = 4$
7. $2x + 3y = 13,\ x + y = 5$
8. 3 pens and 2 pads cost £7; 1 pen and 2 pads cost £5. Find each price.
9. Is $x + y = 3$, $2x + 2y = 6$ one answer, none, or endless?
10. Compounds 3: how much of the 80 g/L and 25 g/L solutions make 70 ml at 45 g/L?

6. $x = 7$, $y = 3$ 7. $x = 2$, $y = 3$ 8. pen £1, pad £2 9. endless (same line)
10. about 25.5 ml and 44.5 ml

Quadratics (Δ first)

11. $x^2 = 49$
12. $x^2 - 5x + 6 = 0$
13. $x^2 + 2x - 8 = 0$
14. $x^2 = 7x$
15. Complete the square: $x^2 + 8x + 10$
16. $x^2 + 4x + 1 = 0$ (2 d.p.)

11. ± 7 12. $\Delta = 1$: $x = 2$ or 3 13. $\Delta = 36$: $x = 2$ or -4 14. $x = 0$ or 7 15. $(x + 4)^2 - 6$
16. $\Delta = 12$: -0.27 or -3.73

Methods, words and inequalities

17. Use substitution: $y = 3x - 1$ and $2x + y = 9$
18. Use elimination (scale both): $2x + 3y = 8$, $3x + 2y = 7$
19. Solve $(x + 2)(x - 3) = 6$
20. Complete the square: $3x^2 - 12x + 5$
21. Solve $7 - 3x > 1$ and show it on a number line
22. A rectangle is 2 m longer than it is wide, with area 48 m². Width?
23. Solve $2/x + x = 3$

17. $x = 2$, $y = 5$ 18. $x = 1$, $y = 2$ 19. $x^2 - x - 12 = 0$: $x = 4$ or -3 20. $3(x - 2)^2 - 7$
21. $x < 2$ (open circle) 22. 6 m (reject -8) 23. $x = 1$ or 2

Find the unknown

24. Find k so that $x^2 + kx + 16 = 0$ has equal roots.
25. One root of $x^2 + kx - 12 = 0$ is 3. Find k and the other root.
26. For which k does $x^2 + 6x + k = 0$ have no real roots?
27. $x = -2$ is a root of $2x^2 + 5x + k = 0$. Find k and the other root.
28. $ax + 2y = 11$ and $x - y = 2$ have the solution $x = 3$. Find y and a.
29. $kx^2 + 4x + 1 = 0$ has equal roots. Find k.

24. $k = \pm 8$ ($k^2 - 64 = 0$) 25. $k = 1$, other root -4 ($9 + 3k - 12 = 0$) 26. $k > 9$ ($36 - 4k < 0$)
27. $k = 2$, other root $-1/2$ ($8 - 10 + k = 0$) 28. $y = 1$, $a = 3$ ($3a + 2 = 11$)
29. $k = 4$ ($16 - 4k = 0$)